IV. Submersions, Immersions, and Embeddings & (Embedded submanifold)

1. Maps of Constant ⇔ Rank

Def IV 1.1. Suppose $F: M \to N$ smooth and $p \in M$. Then the rank of F at p is the rank of $dF_p: T_pM \to T_{F(p)}N$

Rmk. the rank of dF_p equals to the rank of $J(\psi \circ F \circ \varphi^{-1})$ and equals to $\dim \operatorname{Im} dF_p$ in $T_{F(p)}N$

Indeed, $T_pM \xrightarrow{dF_p} T_pN$ where all maps are bijective. and $\operatorname{rank}(J(\psi \circ F \circ \varphi^{-1})) = \operatorname{rank}\, d(\psi \circ F \circ \varphi^{-1})_p$

$d\varphi_p \downarrow$ $\qquad \downarrow d\psi_p$

$\mathbb{R}^m \xrightarrow{d(\psi \circ F \circ \varphi^{-1})} \mathbb{R}^n$

Def IV. 1.2. (1) If F has same constant rank r, then we call rank $F = r$.

(2) If dF is surjective, namely, rank $dF = n$ $(m \geq n)$, then we call F is a smooth submersion (淹入).

(3) If dF is injective, rank $dF = m$ $(m \leq n)$, then we call F is a smooth immersion (浸入)

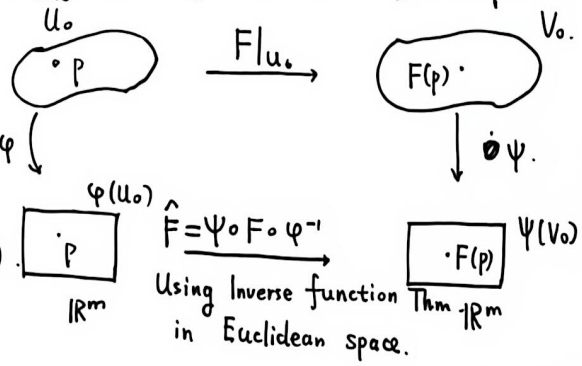
(4) If F is injective, and dF is injective, then we call F is a injective immersion (单浸入)

(5) If F is a smooth immersion, and M and $F(M)$ (with the subspace topology from N) are homeomorphic, then we call F is a smooth embedding (嵌入).

(6). If $\forall p \in M$, $\exists$ neighbourhood U. s.t. $F(U)$ open in N and $F|_U: U \to F(U)$ is a diffeomorphis then we call F is a local diffeomorphism.

Thm IV. 1.3. (Inverse functions Theorem). M, N are smooth manifolds, and $F: M \to N$ smooth. If $p \in M$ s.t. dF_p is invertible, then $\exists$ connected U_0 containing p, V_0 containing $F(p)$ s.t. $F|_{U_0}: U_0 \to V_0$ is a diffeomorphism.

pf (Hint):



where $d\hat{F} = d\psi_{F(p)} \circ dF_p \circ d(\varphi^{-1})_0$ is non-singular

Thm IV. 1.4. (Rank Theorem) Let $F: M \to N$ smooth and rank $F = r$. $\forall p \in M$. $\exists (U, \varphi)$ centered at and (V, ψ) centered at $F(p)$ s.t. $F(U) \subseteq V$., in which F has a coordinate representation

$$\hat{F}(x^1, \dots, x^r, x^{r+1}, \dots, x^m) = (x^1, \dots, x^r, 0, \dots, 0).$$

pf: W.L.O.G. let $M = \mathbb{R}^m$. $N = \mathbb{R}^n$. , $p = (0, \dots, 0)$.

W.L.O.G. $\operatorname{rank}\left(\frac{\partial f^i}{\partial x^j}\right)_{1 \leq i, j \leq r} = r$

~~Def $g: \mathbb{R}^n \to \mathbb{R}^{n-r}$ $(x^1, \dots, x^n) \mapsto (f^{r+1}(x), \dots, f^n(x))$~~

~~Then $J(g)(x) = \left(\frac{\partial f^i}{\partial x^j}\right)_{r+1 \le i,j \le n}$~~ let $(x,y) := (x^1, \dots, x^r, y^1, \dots, y^{n-r}) \in \mathbb{R}^n$

$(v,w) = (v^1, \dots, v^r, w^1, \dots, w^{n-r}) \in \mathbb{R}^n$

$F(x,y) = (Q(x,y), R(x,y))$ where $\left(\frac{\partial Q^i}{\partial x^j}\right)$ non singular

$Q: \mathbb{R}^m \to \mathbb{R}^r$

Define $\varphi: \mathbb{R}^m \to \mathbb{R}^m$ $(x,y) \mapsto (Q(x,y), y)$

where $d\varphi_{(0,0)} = \begin{pmatrix} \frac{\partial Q^i}{\partial x^j}(0,0) & \frac{\partial Q^i}{\partial y^j}(0,0) \\ 0 & I_{m-r,m-r} \end{pmatrix}$ is non-singular

By inverse function theorem, $\exists$ U_0 containing $(0,0) \subseteq \mathbb{R}^m$, $\tilde{U}_0$ containing $\varphi(0,0) \subseteq \mathbb{R}^m$

s.t. $\varphi: U_0 \to \varphi(U_0)$ is a diffeomorphism. Let $\psi = \varphi^{-1}: \varphi(U_0) \to U_0$. denote the $\varphi(U_0) = \tilde{U}_0$

Define $V_0 \subseteq \mathbb{R}^m := \{(v,w) \in V_0 \mid (v,0) \in \tilde{U}_0\}$. $\psi(x,y) := (A(x,y), B(x,y))$

Then $F \circ \psi: \tilde{U}_0 \subseteq \mathbb{R}^m \to V_0$.

$(x^1, \dots, x^r, x^{r+1}, \dots, x^m) \mapsto F(A(x,y), y) = (x, \tilde{R}(x,y))$

where $\tilde{R}(x,y) = R(A(x,y), y)$ $\quad d(F \circ \psi)(x,y) = \begin{pmatrix} I_r & 0 \\ * & \frac{\partial \tilde{R}^i}{\partial y^j}(x,y) \end{pmatrix}$ has the rank r.

$\Rightarrow \frac{\partial \tilde{R}^i}{\partial y^j} = 0$ on $\tilde{U}_0$ (W.L.O.G let $\tilde{U}_0$ be a cube) $\Rightarrow \tilde{R}$ is independent of $(y^1, \dots, y^{m-r})$

$\Rightarrow \underline{\tilde{R}(x,y) = \tilde{R}(x,0)} \overset{S(x)}{=\!=\!=} \quad (**)$

Next define $\phi: V_0 \to \mathbb{R}^n$ $(v,w) \mapsto (v, w - S(v))$ be a isomorphism. $(\phi^{-1}(s,t) = (s, t+S(s))$

Then $\phi \circ F \circ \psi(x,y) = \phi(x, S(x)) = (x,0)$ $\square$

Rmk. ① The most important part of the proof is the construction of $\varphi: \mathbb{R}^m \to \mathbb{R}^m$ $(x,y) \mapsto (Q(x,y), y)$

Suppose

② M is connected, $F: M \to N$ smooth. Then TFAE: (a) $\forall p \in M$, $\exists$ (U,φ) containing p, (V,ψ) containing $F(p)$ s.t. the coordinate representation is linear.

(b) F has constant rank r.

③. Next, we show a global theorem to make it easy to identify various "mersions".

Thm IV.1.5. (Global rank theorem) M, N be smooth, and $F: M \to N$ be smooth with constant rank.

(a). F surjective $\Rightarrow$ F smooth submersion

(b) F injective $\Rightarrow$ F smooth immersion

(c) F bijective $\Rightarrow$ F diffeomorphism.

Pf: Let $m = \dim M$, $n = \dim N$. $r = \operatorname{rank} F$.

(a) Assume that $r < n$. by (IV.1.4), we have $\forall p \in M$, $\exists (U, \varphi)$ containing p, (V, ψ) containing $F(p)$, s.t. $F(x^1, \dots, x^m) = (x^1, \dots, x^r, 0, \dots, 0)$, $F(U) \subseteq V$.

W.L.O.G let $\bar{U}$ be a regular ball, s.t. $F(\bar{U}) \subseteq V$

Then $F(\bar{U}) \subseteq \{y \in V \mid y^{r+1} = \dots = y^n = 0\}$. is compact, hence closed and contains no open subset in V. Therefore, $F(\bar{U})$ is nowhere dense in V !!!

Now choose countably $\{(U_i, \varphi_i)\}$ containing M, with $\{(V_i, \psi_i)\}$ covering $F(M)$

$\Rightarrow F(M) = \bigcup_{n \geq 1} F(\bar{U}_i)$ $F(\bar{U}_i)$ be nowhere dense.

By Baire category theorem, $F(M)^\circ = \phi$. $\Rightarrow$ F isn't surjective, contradiction!

(b) Assume that $r < m$. ..., we have $F(x^1, \dots, x^m) = (x^1, \dots, x^r, \underbrace{0, \dots 0}_{m}, \underbrace{\quad}_{(n-m) \text{ many}})$

$\Rightarrow F(0, \dots, 0, \varepsilon) = F(0, \dots, 0)$. for ε sufficiently small ε.

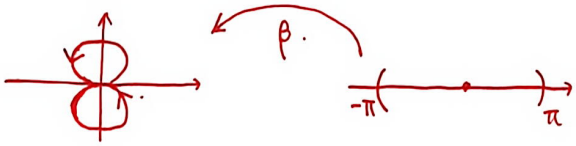
Contradiction!

(c). F is both a immersion and a submersion. hence local diffeomorphism. $\overset{\text{bijective}}{\Rightarrow}$ diffeomorphism $\square$

2. Embeddings, submanifold.

Two failed examples where F can be an injective immersion, BUT not the embedding!

① $\beta: (-\pi, \pi) \to \mathbb{R}^2$ $\quad t \mapsto (\sin 2t, \sin t)$ (lemniscate) or written as $x^2 = 4y^2(1-y^2)$



② Let $\mathbf{T}^2 = S^1 \times S^1 \subseteq \mathbb{C}^2$. be the torus. $\gamma: \mathbb{R} \to \mathbf{T}^2$ $\quad t \mapsto (e^{2\pi i t}, e^{2\pi i \alpha t})$

where $\alpha \in \mathbb{R} \setminus \mathbb{Q}$.

Thm IV.2.1. (Local Embedding Thm). M, N smooth, and $F: M \to N$ smooth. Then F is a smooth immersion iff $\forall p \in M$. $\exists U$ containing p, s.t. $F|_U$ is a smooth embedding.

pf: $\Leftarrow$: is obvious

$\Rightarrow$: $\forall p \in M$. $\exists (U, \varphi)$ containing p, (V, ϕ) containing $F(p)$ s.t. $F(x^1, \dots, x^m) = (x^1, \dots, x^m, 0 \dots 0)$

hence injective

$\exists U_0$ paracompact. s.t. $\overline{U_0} \subseteq U$.

Then $F|_{\bar{U}_0}$ is an injective continuous map with compact domain $\Rightarrow$ $F|_{\overline{U_0}}$ is a topological embedding $\Rightarrow$. $F|_{U_0}$ is a smooth embedding $\square$